Existence of a strong Diophantine quadruple

Problèmes ouverts — A. Dujella · 2026

Problem 1.15 (Dujella, 2026). Is there any strong Diophantine quadruple, i.e. a set of four non-zero rationals $\{a_1, a_2, a_3, a_4\}$ such that $a_i a_j + 1$ are perfect squares for all $1 \leqslant i, j \leqslant 4$?

Remark. There are examples of "almost" strong Diophantine quadruples, where all conditions for a strong quadruple are satisfied except that the condition that $a_3 a_4 + 1$ is a square is not satisfied, e.g. $\{\frac{140}{51}, \frac{2223}{3046}, \frac{278817}{33856}, \frac{3182740}{17661}\}$ (Dujella-Petričević, Experiment. Math. **17** (2008), 83–89), but it is not known whether there are infinitely many such examples.